

String Diagrams, Quantum Mechanics, and AI

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Processes and Compositionality

- We can view things in terms of their processes
- This is pertinent in quantum mechanics
- AI architectures are modular and also in need of a process oriented language

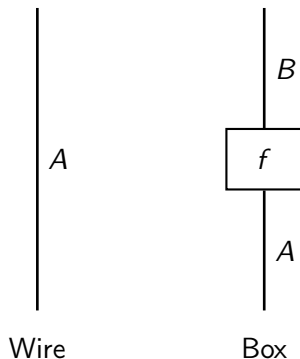
Category Theory

- We have objects: $A, B, C \dots \in Ob(\mathcal{C})$
- We have arrows:

$$f : A \rightarrow B, g : B \rightarrow C, \dots \in Mor(\mathcal{C})$$

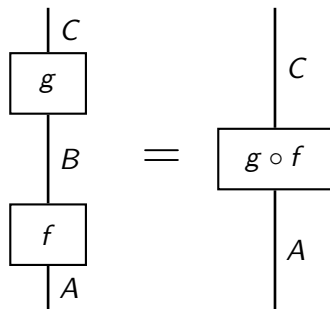
- The arrows can compose: $g \circ f : A \rightarrow C$
- Composition is closed, associative, and unital

Graphical Syntax: Boxes and Wires



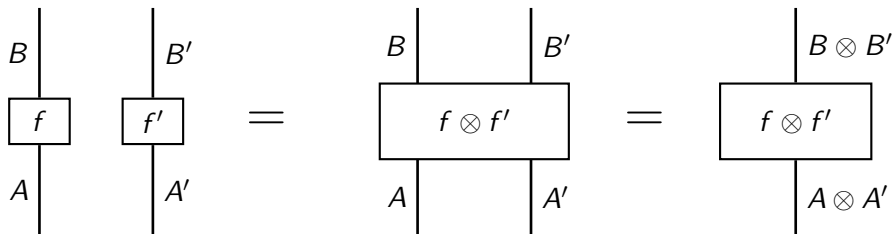
- Wires correspond to objects $A, B, C, \dots \in Ob(\mathcal{C})$
- Boxes correspond to morphisms $f : A \rightarrow B, g : B \rightarrow C, \dots \in Mor(\mathcal{C})$

Graphical Syntax: Composing Boxes



- The graphical composition $g \circ f$ of two morphisms $f, g \in \text{Mor}(\mathcal{C})$

Graphical Syntax: Parallel Processes



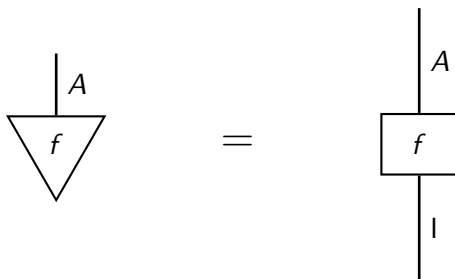
- This is a construction in monoidal categories

Graphical Syntax: Identity Object



- All monoidal categories have an identity object I that satisfies several properties
- We have $\lambda_A : I \otimes A \rightarrow A$
- We also have this property for right-hand identities

Graphical Syntax: States



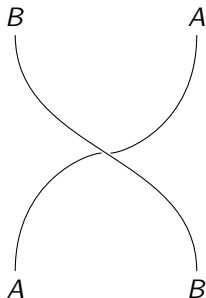
- **States:** $f : I \rightarrow A$

Graphical Syntax: Costates



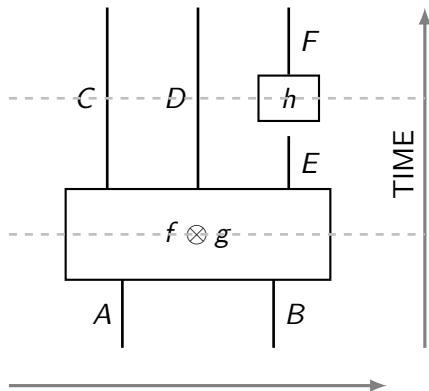
- **Costates** are morphisms $f : A \rightarrow I$
- These are often referred to as **effects** in quantum mechanics

Graphical Syntax: Braiding Processes



- For a braiding monoidal category, we have the natural isomorphism $\sigma_{A,B} : A \otimes B \rightarrow B \otimes A$
- For symmetric monoidal categories, we have $\sigma_{A,B} \circ \sigma_{B,A} = Id_{A \otimes B}$

The Arrow of Time



PARALLEL / SIMULTANEOUS PROCESSES

- Typically quantum physics, AI, and other fields rely on static derivations
- One of the powerful views that category theory provides is the ability to see the arrow of time

String Diagrams Satisfy Coherence Properties

Proof by trust me bro.

Normal Quantum Mechanics

- Our wave states correspond to vectors in a Hilbert space $\mathcal{H}$ over $\mathbb{C}$
- Composite systems correspond to tensored Hilbert spaces:
 $\mathcal{H}_A \otimes \mathcal{H}_B \otimes \dots$
- Born's Rule: $|\langle \phi | \psi \rangle|^2 = P(\phi | \psi)$
- Operators are bounded linear maps

How to View Quantum Mechanics Categorically

- We use the dagger compact monoidal category for quantum mechanics¹
- Objects are Hilbert spaces
- Morphisms are bounded linear maps
- Identity object is $\mathbb{C}$
- A dagger functor $\dagger : \text{Hilb} \rightarrow \text{Hilb}$
 - ▶ $(g \circ f)^\dagger = f^\dagger \circ g^\dagger$
 - ▶ $id_A^\dagger = id_A$
 - ▶ $(f^\dagger)^\dagger = f$
- $|\psi\rangle : \mathcal{H}_A \rightarrow \mathbb{C}$
- $\langle\psi| : \mathbb{C} \rightarrow \mathcal{H}_A$

Quantum Circuits

- The quantum circuits seen in quantum computing naturally satisfy string diagrams formalism (albeit slightly different syntax)

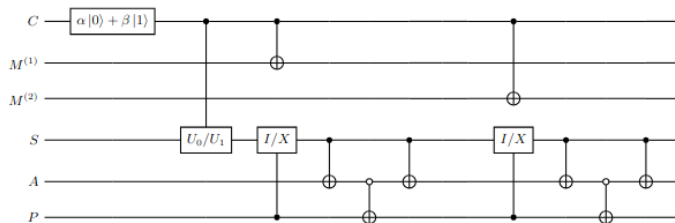


FIG. 12: Quantum circuit for the 2-iteration QDM that implements outcome-driven oracle-based adaptive policy update, which helps steer the system register S to the desired target state $|1\rangle_S$, regardless of the initial branch-dependent “push” performed by control C .

Figure: Taken from *Galiautdinov 2025*²

Normal AI

- AI models consist of parameters and transformations acting on these parameters

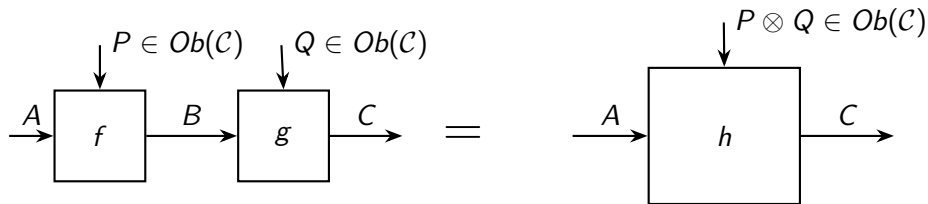
Normal Multilayer Perceptron

Definition (MLP)

Multilayer Perceptron: $f_{\theta}(x) : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^m$

$f(x)_{\theta} = SM \circ \mathcal{A}_1 \circ \sigma_1 \circ \dots \circ \mathcal{A}_L \circ \sigma_L$, where σ_i is a non-linear transformation, $\mathcal{A}_i$ is an affine transformation, and SM is softmax

Categorical AI with Data, Parameters, and Transformations



- The category $\mathbf{Para}(\mathcal{C})^{3,4}$
- Horizontal wires are data-types, vertical wires are parameters
- Morphisms $f : A \times P \rightarrow B$, $g : B \times Q \rightarrow C$
- NOTE: I rotated our time axis 90 degrees

Categorical AI: Reparameterizations

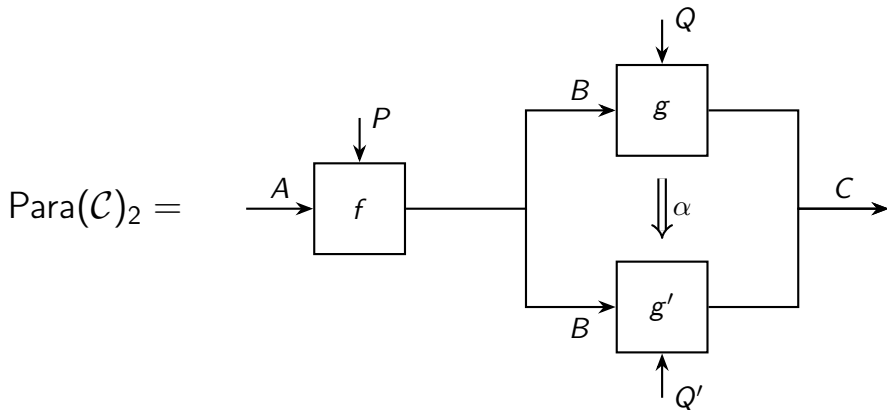


Figure: α is a 2-morphism between our morphisms g and g' (the double arrow). If g and g' are parametric mappings, then α is the reparameterization from g to g' .

String Diagrams in Quantum Mechanics are Complete

Any mathematics done in quantum mechanics can be done diagrammatically (this was proved in 2019)⁵

Alternative to Logic and Typing

- Enforces typing (Howard-Lambek Correspondence)
- Correspondence to intuitionistic linear logic (Curry-Lambek Correspondence)

Conclusion

- String diagrams give fast, intuition-based, rigorous derivations
- They allow us to see the forest through the trees
- They instantiate proper typing
- Support modeling for AI and quantum mechanics

Bibliography

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